Mini Project: Student Performance Analysis

🔷 Step 1: Create Dataset (CSV file)

Create a file called students.csv and paste this:

```
Name,Math,Science,English
Amit,78,85,88
Bhavana,92,90,94
Charan,67,70,72
Divya,88,76,85
Esha,95,92,91
Farhan,55,60,58
Gita,80,83,79
Hari,72,68,74
Isha,89,91,90
John,60,65,62
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("students.csv")

# Display data
print("Dataset:\n", df)

# Basic info
print("\nInfo:")
print(df.info())

# Statistical summary
print("\nSummary:")
print(df.describe())
```

```
# Add new column: Average marks
df['Average'] = np.mean(df[['Math', 'Science',
'English']], axis=1)

# Add Grade column
def grade(avg):
    if avg >= 90:
        return 'A'
    elif avg >= 75:
        return 'B'
    elif avg >= 60:
        return 'C'
    else:
        return 'D'

df['Grade'] = df['Average'].apply(grade)

print("\nUpdated Dataset:\n", df)
```

```
# ----------------------------
# 📊 Visualization
# ----------------------------

# 1. Bar chart - Average marks
plt.figure()
plt.bar(df['Name'], df['Average'])
plt.title("Average Marks of Students")
plt.xlabel("Students")
plt.ylabel("Average Marks")
plt.xticks(rotation=45)
plt.show()
```

```
# 2. Pie chart - Grade distribution
plt.figure()
df['Grade'].value_counts().plot.pie(autopct='%1.1f%%')
plt.title("Grade Distribution")
plt.ylabel('')
plt.show()

# 3. Line plot - Subject comparison
plt.figure()
plt.plot(df['Name'], df['Math'], label='Math')
plt.plot(df['Name'], df['Science'], label='Science')
plt.plot(df['Name'], df['English'], label='English')
plt.legend()
plt.title("Subject-wise Marks")
plt.xticks(rotation=45)
plt.show()
```